IM3570 Sample Program Instructions

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I. Sample Program Instructions

Section 1.01 Launching the sample program

Double click to open the "HIOKI IM3570 LabVIEW Driver Release V.1.0.llb" file downloaded from download section of HIOKI IM3570 section.

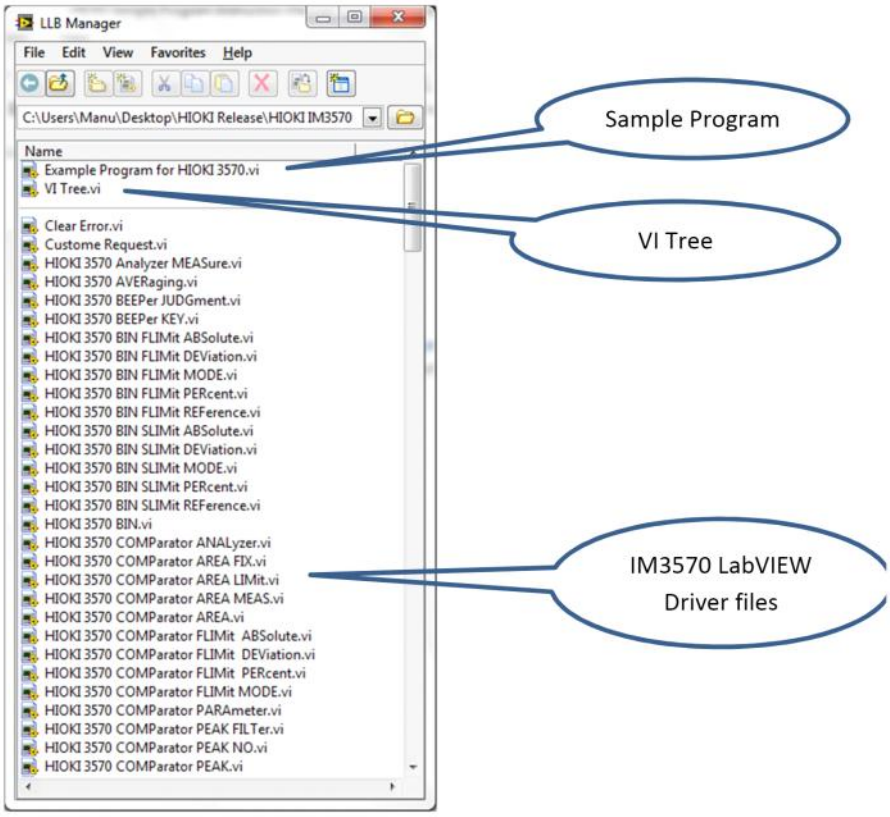


Figure 1: LLB File Contents

Figure 1 show the position of sample program when HIOKI IM3570 LLB file is opened. Double-click the same to open the sample program. A LabVIEW installation with version above 8 must be installed in the PC to view the same.

"VI Tree" VI can be used to view the entire instrument driver hierarchy. The VI Tree VI is a no executable VI that illustrates the functional structure of the instrument driver.

Section 1.02 Using the sample program

Double-Click the "Example Program for HIOKI 3570.vi" to open the sample program from provided LLB file. The application shown like Figure 2 pops-up when the following sample program is launched.

Example Program for HIOKI 3570.vi

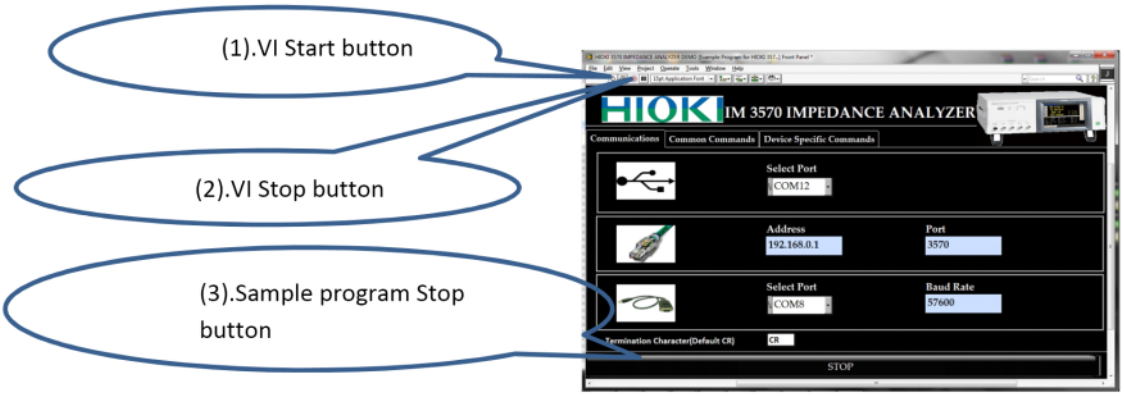


Figure 2: Sample Program before pressing VI start button

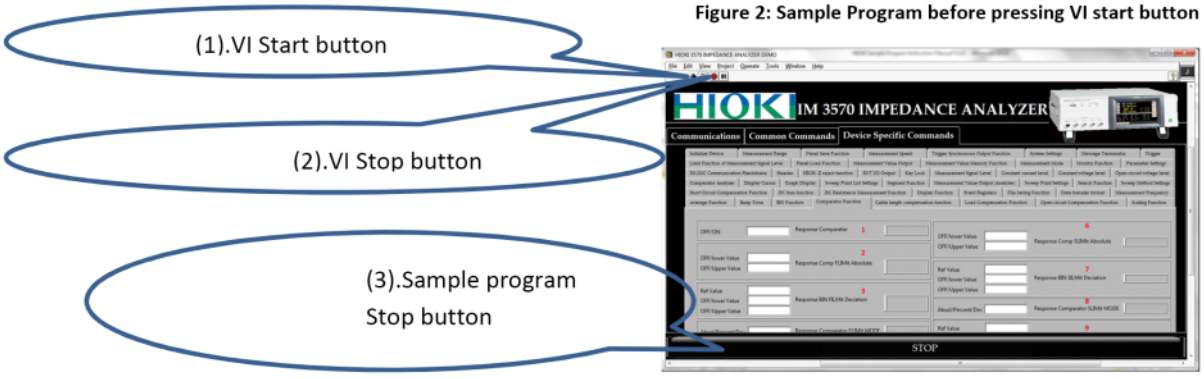


Figure 3: Sample Program after pressing VI start button

(1).VI Start button starts the sample program. (3). Sample program Stop button stops the application.

(2). VI stop button should be used if (3). Sample program Stop button cannot stop the program.

II. Example Program for HIOKI 3570.vi

Example Program for HIOKI 3570.vi is subdivided into 3 main screens.

1. Communications Screen
2. Common Commands Screen
3. Device Specific Commands

Section 2.01 Communication Screen

Communication screen as in Figure 4 in "Example Program for HIOKI 3570.vi" is used for setting communication interface between IM 3570 and PC. There are 3 ways to establish the communication interface.

(a) Selecting USB Interface

IM3570 can be connected to PC via USB interface. Prior to connecting the device to PC, please make sure that USB interface is selected in IM 3570. This can be verified in device by checking following path.

SYS>>IF>>USB

More details on setting the same can be found from IM3570 User Manual.

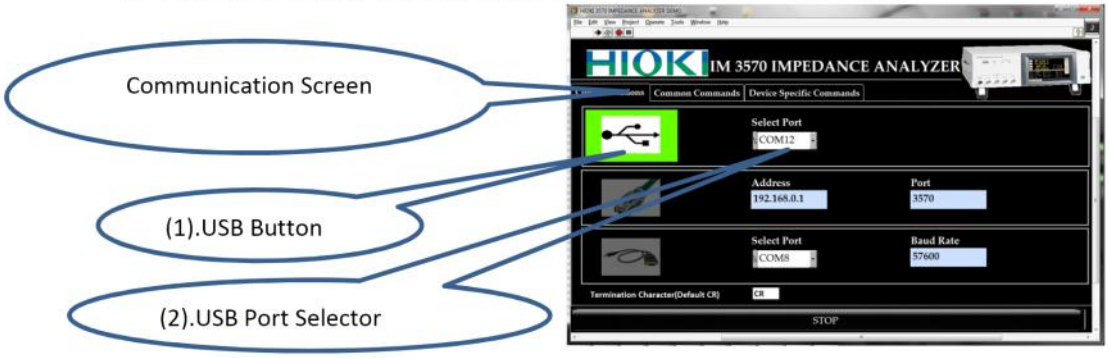


Figure 4: Selecting USB Interface for Communication

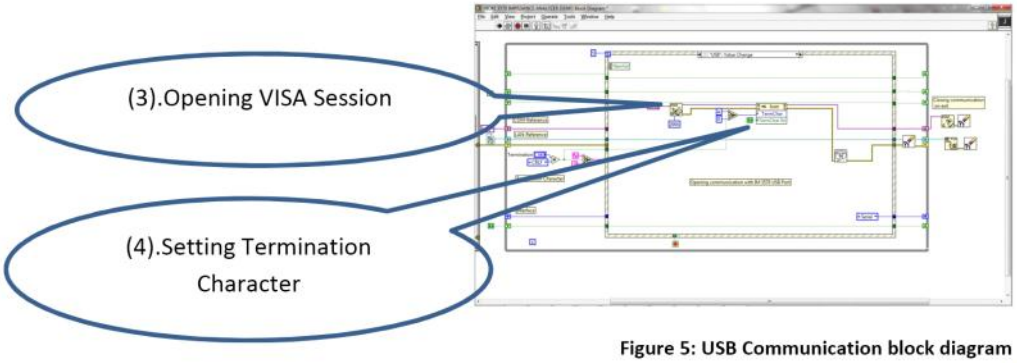


Figure 5: USB Communication block diagram

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To select USB Communication,

1. Select the COM port (can be found from device manager of PC) where the instrument connected. This can be achieved using the (2). USB Port Selector button shown in Figure 4.
2. Press "(1).USB Button" to initialize communication between PC and IM 3570 as shown in Figure 4.
3. Press Keyboard "Ctrl+E" button from application to view the LabVIEW block diagram as shown in Figure 5.
4. **Step and 1 and 2 must be done prior to pressing the run button.**

(b) Selecting LAN interface

IM3570 can be connected to PC via LAN interface. Prior to connecting the device to PC, please make sure that LAN interface is selected in IM 3570. This can be verified in device by checking following path.

SYS>>IF>>LAN

Note the IP address and port address to enter in the application.

More details on setting the same can be found from IM3570 User Manual.

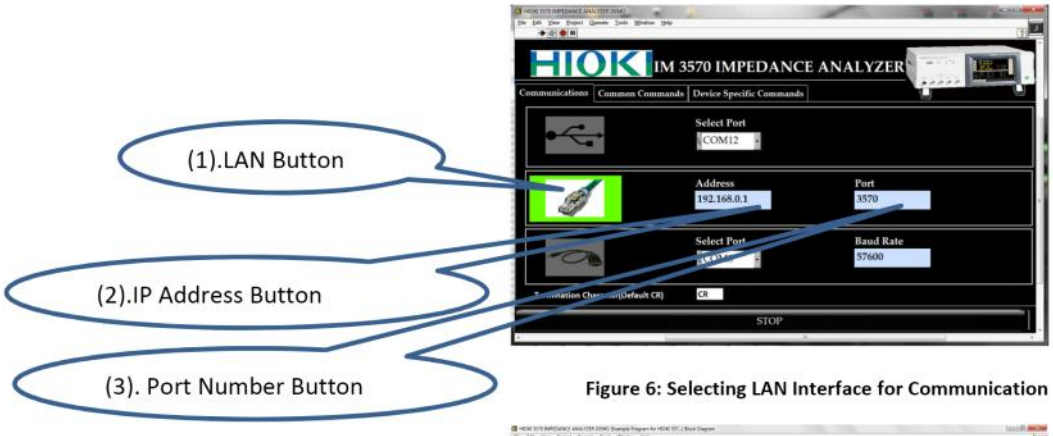


Figure 6: Selecting LAN Interface for Communication

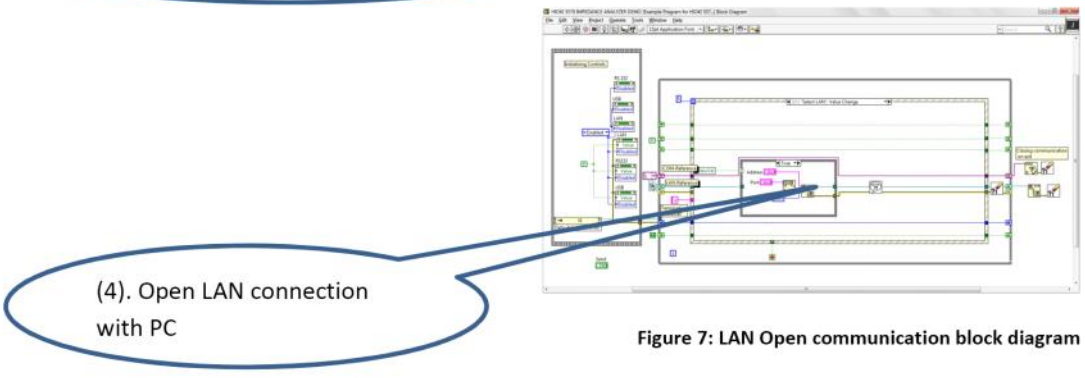


Figure 7: LAN Open communication block diagram

To select LAN Communication,

1. Enter the IP address in "(2).IP Address Button" as in Figure 6.
2. Enter the port number in "(3).Port Number Button" as in Figure 6.
3. Press "(1).LAN Button" to initialize communication between PC and IM 3570 as in Figure 6.
4. Press Keyboard "Ctrl+E" button from application to view the LabVIEW block diagram as shown in Figure 7.
5. **Perform step 1, 2 and 3 prior to pressing the VI start button.**

(c) Selecting RS-232 interface

IM3570 can be connected to PC via RS-232 interface. Prior to connecting the device to PC, please make sure that RS-232 interface is selected in IM 3570. This can be verified in device by checking following path.

SYS>>IF>>RS232

Note the serial port settings in IM 3570 to enter in the application.

More details on setting the same can be found from IM 3570 User Manual.

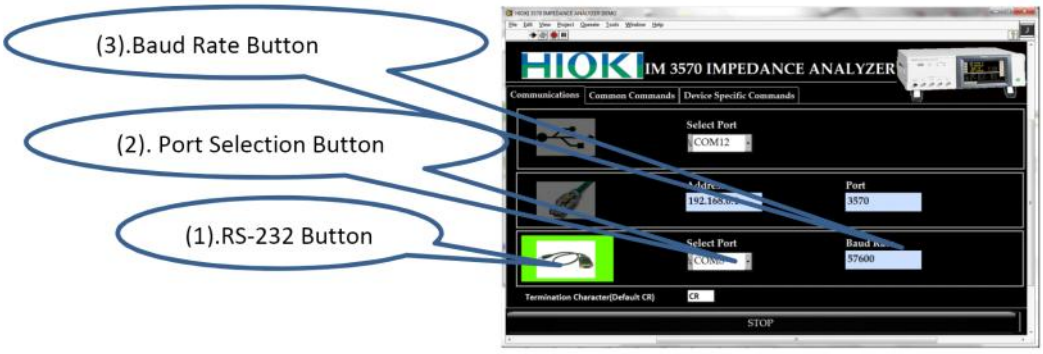


Figure 8: Selecting RS-232 Interface for Communication

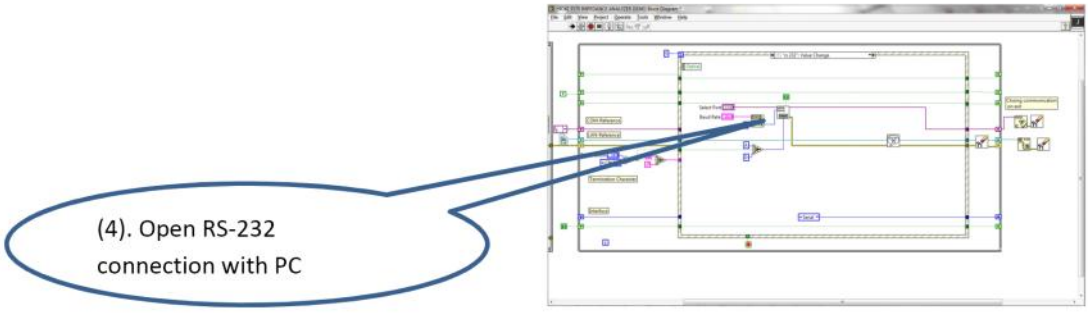


Figure 9: Opening communication with RS-232 Block Diagram

To select RS-232 Communication,

1. Select the port address in "(2).Port Selection Button" as in Figure 8.
2. Enter the Baud Rate in "(3).Baud Rate Button" as in Figure 8.
3. Press "(1).RS-232 Button" to initialize communication between PC and IM 3570 as in Figure 8.
4. Press Keyboard "Ctrl+E" button from application to view the LabVIEW block diagram as shown in Figure 9.
5. **Perform step 1, 2 and 3 prior to pressing the VI start button.**

(d) Selecting Termination Character

The termination character in the device should be set in the sample program. More details on setting the same can be found from IM 3570 User Manual.

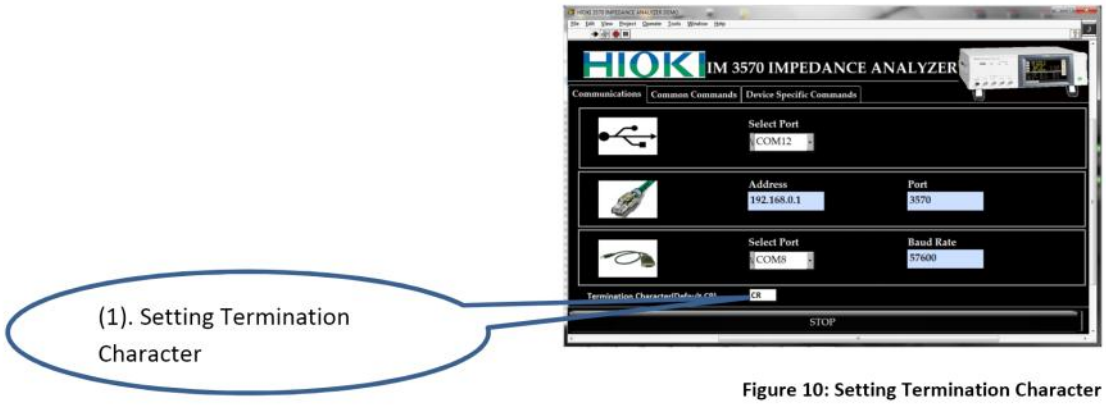


Figure 10: Setting Termination Character

Figure 10 shows the place where user can select termination character of input string coming from the instrument. The available options are CR (Carriage Return) and CRLF (Carriage Return+ Line Feed).

It is mandatory to have the same settings in instrument and sample program.

Section 2.02 Common Commands Screen

Common Commands screen in "Example Program for HIOKI 3570.vi" is used for sending and querying common SCPI communication commands to IM3570 from PC. This screen is specific for items listed in in *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, and Section 4.1.1 Common Commands List*

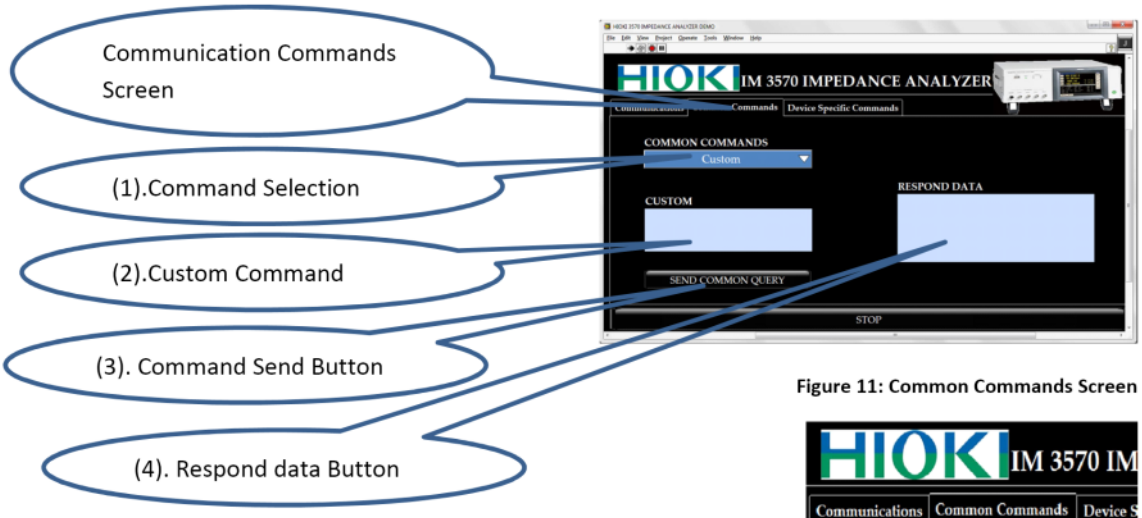


Figure 11: Common Commands Screen

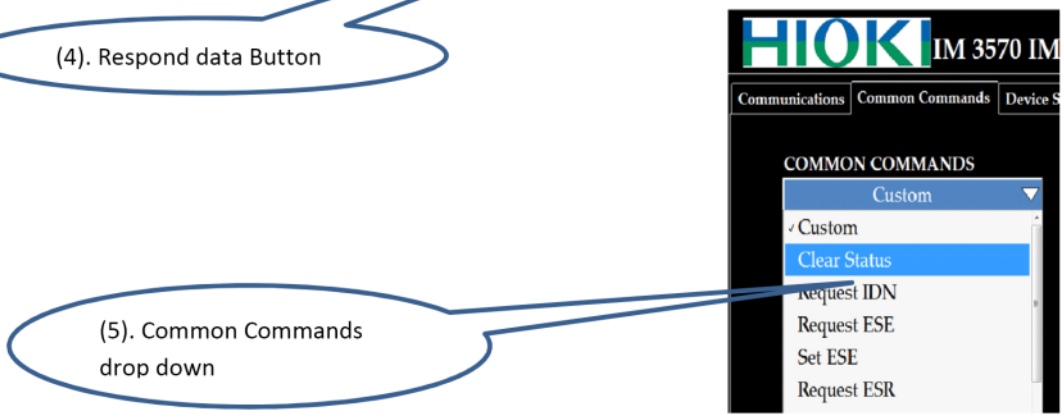


Figure 12: Common Commands

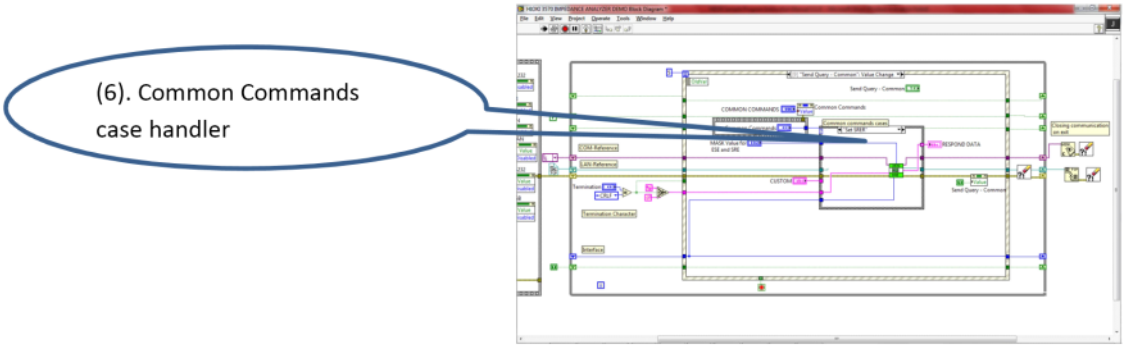


Figure 13: Common Commands block diagram

Following are the major sections of common commands screen:

(1).Command Selection

Figure 10 shows the command selection button for common commands. The available common commands are detailed in *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, Section 4.1.1 Common Commands List*. To query or set a particular command, click this drop down and select the required command. To perform the operation, Press "(3). Command Send Button". If the performed operation expects a response from device, the output response can be found at "Respond data" field, shown at Figure 10. Figure 11 shows a sample drop-down of common commands.

(2).Custom Command

If a new command, which is not listed in command selection list, *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, Section 4.1.1 Common Commands List*, select the command as custom command in command selection list and perform the query using "(3). Command Send Button".

(3). Command Send Button

"Command send button" as shown in Figure 10 is used for sending command specified in "(1).Command selection list. Press this button after running the program to perform the operation.

(4). Respond Button

"Respond Button" as shown in Figure 10 displays the received data from instrument as a result of any queries from PC.

(5). Common Commands drop down

Figure 11 displays a sample command selection drop down.

(6). Common Commands case handler

Press Keyboard "Ctrl+E" button from application to view the LabVIEW block diagram as shown in Figure 12. The displayed case structure handles common commands specific queries and commands. Those are the drivers specific for IM 3570 common commands.

Section 2.03 Unique Commands Screen

Unique Commands screen in "Example Program for HIOKI 3570.vi" is used for sending and querying unique SCPI communication commands to IM3570 from PC. This screen is specific for items listed in in *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, Section 4.1.2 Unique Commands List.*

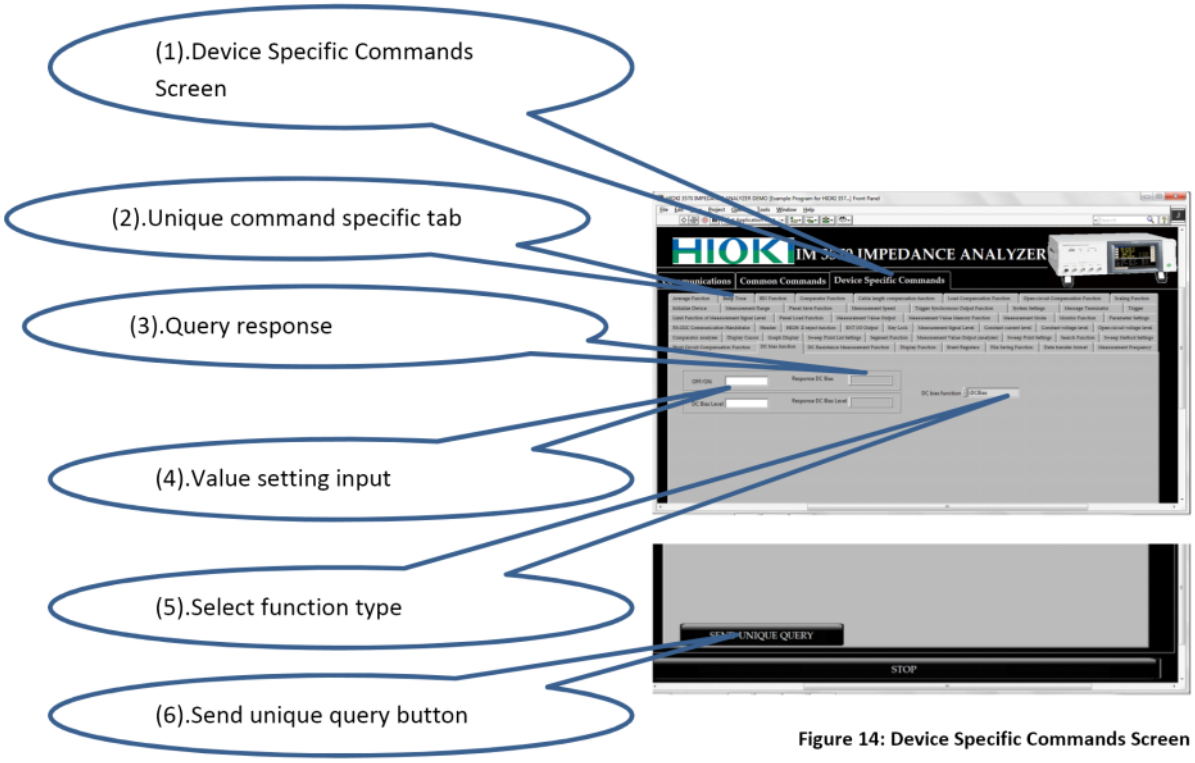


Figure 14: Device Specific Commands Screen

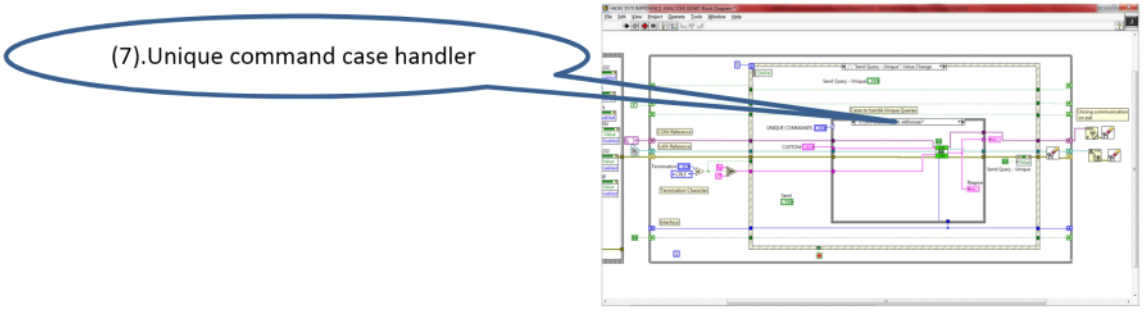


Figure 15: Device Specific commands block diagram

All the commands and queries in unique commands list of IM 3570 can be tested with unique commands screen. The following are the functions of Unique commands screen. The above screen is a sample tab to demonstrate the functionality. The position of these controls may vary in each tab with respect to the number of sub functions available with each command.

(1).Device Specific Commands Screen

Figure 13 screens is similar to common commands screen, except this screen helps to test *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, Section 4.1.2 Unique Commands List session.*

Each tab in device specific commands screen belongs to functionality in unique command list of IM3570 manual. All the sub functions corresponding to that particular function can be accessed by "(5).select function type" control specific for each tab if applicable. It is assumed that the user shall correlate each tabs with *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, Section 4.1.2 Unique Commands List session* to find out which controls and indicators will belong to which function type specified by "(5).select function type" control. Some commands will only provide response values where as some functions needs to be supplied with input parameters in the relevant section. Provide the required input parameters prior to sending the query\command with the help of "(6).Send unique query button".

(2).Unique command specific tab

Figure 13 shows all available tabs with unique command screen. Each tab corresponds to functionality specified in *HIOKI IM3570 Instruction Manual, Chapter 4, Message List, Section 4.1.2 Unique Commands List session.*

(3).Query response

Figure 13 screen allows the user to send query to the IM 3570 by pressing "(6).Send unique query button". All the query commands will respond with a response string from the device and this will be displayed in "Query response" indicators. The position of these indicators will vary with each function tab as number of expected response controls may vary with respect to selected function.

(4).Value setting input

Some of the commands to be sent to the device may require arguments as part of the function and these arguments will be supplied with "Value setting input". The position of these controls will vary with each function tab as number of supplied arguments may vary with respect to selected function.

(5).Select function type

Most of the function tab may contain more than one operation (Command\Query) and this will be supplied with "Select function type control" as shown in Figure 13.

(6).Send unique query button

Figure 13 shows the position of "Send Query" button, which is at the bottom of the screen. **Press this button only after setting the right parameters in all other controls of the selected tab. Any wrong commands will create error in instrument indicated by "Beep" Sound.**

(7).Unique command case handler

Press "Ctrl+E" button from application to view the LabVIEW block diagram as shown in Figure 14.

The case structure shown in Figure 14 handles all the device specific queries and setting drivers for IM 3570.